

2

PARSING

2.1 CD - Grammar

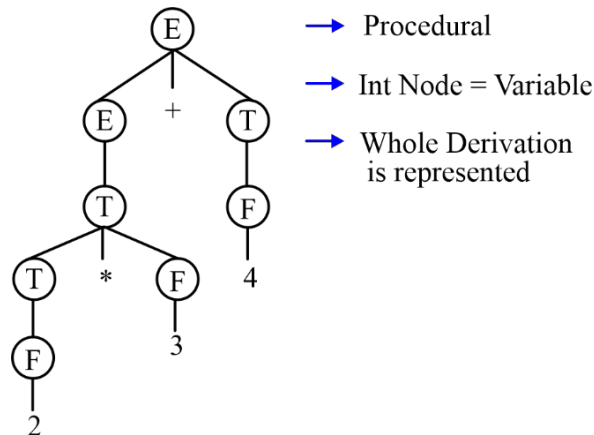
- In compiler we only use: Type – 2 (CFG) and Type – 3 Regular grammar.
- Compiler = Program of Grammar
- Compiler = Membership algorithm
- Every programming Language is Context Sensitive Grammar (Context Sensitive Language)

2.1.1 Parse Tree and Syntax Tree

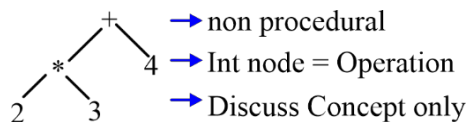
G: $E \rightarrow E + T / T$ $E \rightarrow E + T \rightarrow T + T \rightarrow T * F + T \rightarrow F * F + T \rightarrow 2 * F + T \rightarrow 2 * 3 + T \rightarrow 2 * 3 + F$
 $T \rightarrow T * F / F$

$E \rightarrow 2 * 3 + 4$

Parse Tree:

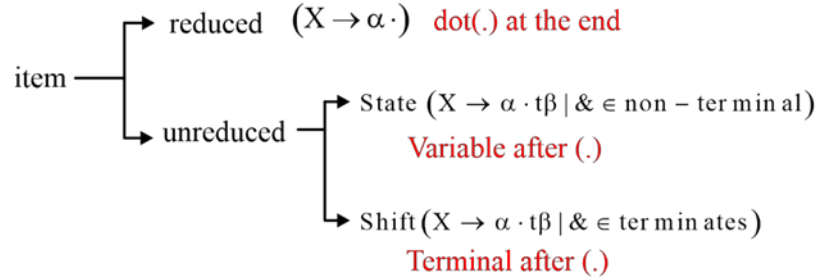


Syntax Tree:



- To check to priority / Associativity:
Randomly derive till you have enough operators, then check which one is done first.
- If priority of 2 operators is same and both Left and Right associative → **Ambiguous Grammar** [Useless]

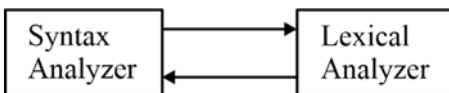
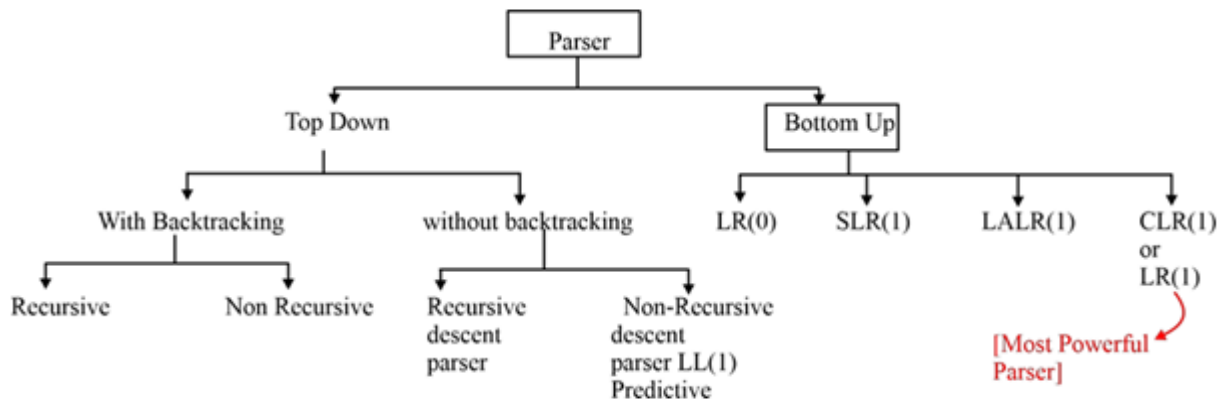
2.1.2 Types of Parsers in Bottom Up: [CD - Parser]



2.2 CD-Syntax Analysis / Parsing

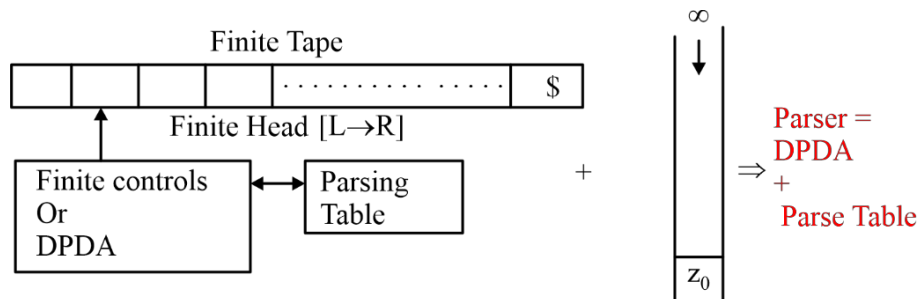
Grammatical Errors are checked by the help of parsers.

- Parsers are basically DPDA



- All of these parsers are table driven.

2.2.1 Mathematical Model of Parser



- Parsers generate Parse Tree, for a given string by the given grammar.